

Kafka Console SOP – Implementation & Troubleshooting (dev-kafka)

Hostname for Console: dev-kafkaconsole.apps.dev.sbiepay.sbi

SOP

1. Purpose

This document defines the standard procedure for deploying Kafka Console, integrating metrics, and troubleshooting issues.

2. Architecture Overview

Kafka Console depends on Kafka Cluster, JMX Exporter, PodMonitor, OpenShift Monitoring, and Console UI.

3. GitLab Structure

deployment/dev/charts/epay_common/templates/

4. Deployment Steps

1. Commit to GitLab
2. Deploy via Helm
3. Verify resources
4. Restart console

4.A. Troubleshooting Table

Metrics issue >Fixed the config

SSL issue >Added CA

Helm issue > Avoided secret conflicts

5. Validation

Check metrics exposure, console config, and logs.

6. Troubleshooting

Common issues include metrics not visible, SSL issues, and Prometheus connectivity problems.

7. Screenshots

All configs in GitLab, deployed via CI/CD.

Kafka-console.yaml

apiVersion: console.streamshub.github.com/v1alpha1

kind: Console

metadata:

name: dev-console

namespace: dev-kafka

spec:

hostname: dev-kafkaconsole.apps.dev.sbiepay.sbi

metricsSources:

- name: kafka-metrics

type: openshift-monitoring

trustStore:

type: PEM

content:

valueFrom:

configMapKeyRef:

name: kube-root-ca.crt

key: ca.crt

kafkaClusters:

- name: dev-kafka-cluster

namespace: dev-kafka

listener: plain

metricsSource: kafka-metrics

properties:

values:

- name: security.protocol
value: PLAINTEXT
- name: connections.max.idle.ms
value: "540000"
- name: reconnect.backoff.ms
value: "1000"
- name: reconnect.backoff.max.ms
value: "10000"

adminProperties:

values:

- name: security.protocol
value: PLAINTEXT
- name: request.timeout.ms
value: "30000"
- name: default.api.timeout.ms
value: "30000"
- name: connections.max.idle.ms
value: "540000"
- name: reconnect.backoff.ms
value: "1000"

- name: reconnect.backoff.max.ms
value: "10000"

consumerProperties:

values:

- name: security.protocol
value: PLAINTEXT
- name: request.timeout.ms
value: "30000"
- name: connections.max.idle.ms
value: "540000"
- name: reconnect.backoff.ms
value: "1000"
- name: reconnect.backoff.max.ms
value: "10000"

producerProperties:

values:

- name: security.protocol
value: PLAINTEXT
- name: request.timeout.ms
value: "30000"
- name: connections.max.idle.ms
value: "540000"
- name: reconnect.backoff.ms
value: "1000"

- name: reconnect.backoff.max.ms

value: "10000"

EPay / DevOps / deployment

main deployment / dev / charts / epay_common / templates / kafka +

Lock Find file Edit Code

 Update file kafka-console.yaml
Suraj Bharat Andhale V1021495 authored 1 hour ago

 e4786789 History

 Code owners Assign users and groups as approvers for specific file changes. [Learn more.](#) Manage branch rules

Name	Last commit	Last update
..		
 configmap.yaml	Update file configmap.yaml	2 hours ago
 kafka-cluster.yaml	Update 3 files	1 week ago
 kafka-connect.yaml	Update 216 files	5 months ago
 kafka-console.yaml	Update file kafka-console.yaml	1 hour ago
 kafka-mirror-maker.yaml	Update 216 files	5 months ago
 kafka-podmonitor.yaml	Update 3 files	1 week ago
 kafka-topics.yaml	Update 216 files	5 months ago
 kafka-user.yaml	Update 216 files	5 months ago

Kafka-metrics.yaml

main deployment / dev / charts / epay_common / files +

Lock Find file Edit Code

 Update 216 files ***
* ISF Devendra Sharma authored 5 months ago

 bbe31715 History

 Code owners Assign users and groups as approvers for specific file changes. [Learn more.](#) Manage branch rules

Name	Last commit	Last update
..		
 kafka-metrics.yaml	Update 216 files	5 months ago

Streams for Apache Kafka Console using Operator :

[Installed Operators](#) > [Operator details](#)



Streams for Apache Kafka Console

3.03-3 provided by Red Hat

★

Actions

DetailsYAMLSubscriptionEventsConsole

Consoles

Show operands in: ☒ All namespaces ☐ Current namespace only

Create Console

Name

Search by name...

Name	Kind	Namespace	Status	Labels	Last updated
 dev-console	Console	 dev-kafka	Condition:  Ready	app.kubernetes.io/managed-by=Helm	 Apr 21, 2026, 4:40 PM

Project: dev-kafka

[Installed Operators](#) > [amp-streams-console/v3.03-3](#) > [Console details](#)



dev-console

DetailsYAMLResourcesEvents

Console overview

Name

dev-console

Namespace

 dev-kafka

Labels

app.kubernetes.io/managed-by=Helm

Edit

Annotations

2 annotations

Created at

 Apr 21, 2026, 4:40 PM

Owner

No owner

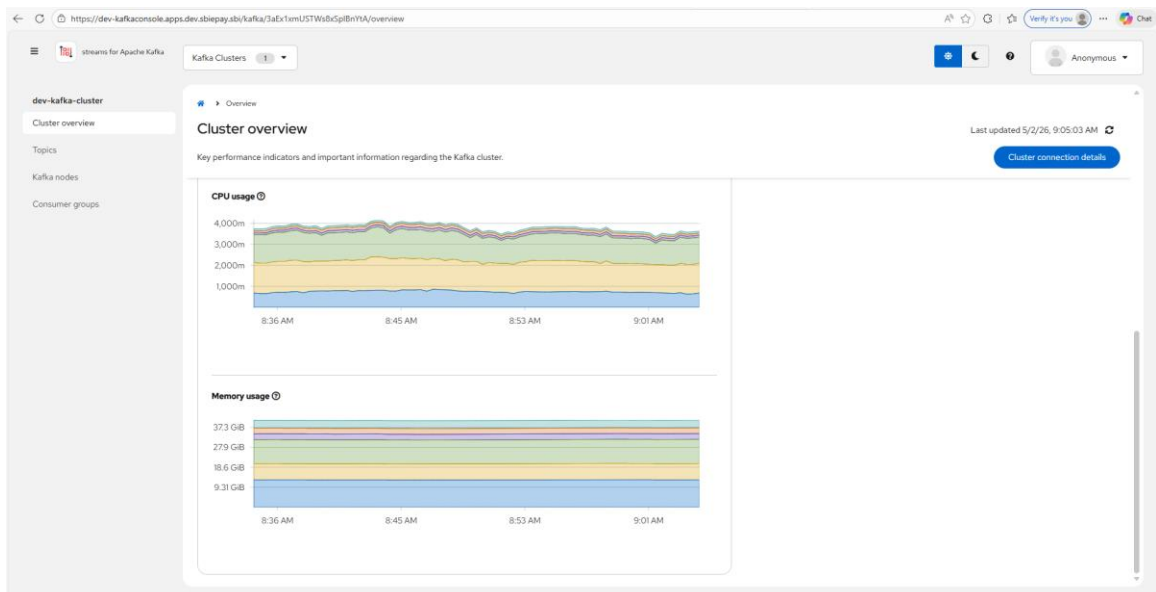
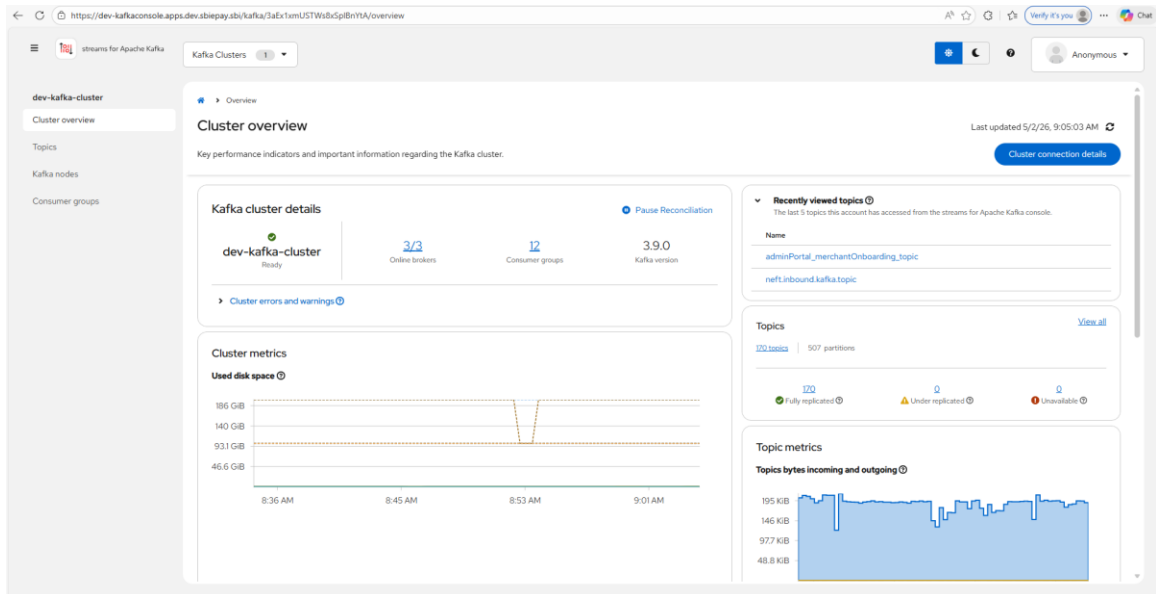
Conditions

Type	Status	Updated	Reason	Message
Ready	True	 May 2, 2026, 1:09 PM	-	All resources ready

Hostname for Console: dev-kafkaconsole.apps.dev.sbiepay.sbi

- All the details related the kafka cluster is visible in the web ui including the Cluster Metrics (Disk, Memory, Cpu Usage).

- All the details related to Kafka Topics are visible in the cluster ui including Graph for throughput (Topics bytes incoming and outgoing) in Topic Metrics section.



All Topic present in dev-kafka cluster :

Name	Status	Partitions	Consumer groups	Storage
Invalid topic name	Fully replicated	1	0	0 B
adminPortal_email_notification_topic	Fully replicated	4	1	0 B
adminPortal_merchantOnboarding_topic	Fully replicated	4	0	0 B
adminPortal_merchantonboarding_topic	Fully replicated	4	1	0 B
adminPortal_sms_notification_topic	Fully replicated	4	1	795 KiB
dev-connect-cluster-configs	Fully replicated	1	0	19.9 KiB
dev-connect-cluster-offsets	Fully replicated	25	0	118 KiB
dev-connect-cluster-status	Fully replicated	5	0	537 KiB
eventtopics	Fully replicated	1	0	0 B
gst_invoice_download_topic	Fully replicated	4	0	0 B
gst_report_status_topic	Fully replicated	4	0	0 B
inb_offline_gateway_pooling_local	Fully replicated	4	1	0 B
local-process-bookingdetails-topic	Fully replicated	4	0	0 B
local-process-bulk-refund-topic	Fully replicated	4	0	0 B
local_adminPortal_email_notification_topic	Fully replicated	4	1	0 B
local_adminPortal_merchantOnboarding_topic	Fully replicated	4	0	0 B

Name	Status	Partitions	Consumer groups	Storage
adminPortal_merchantonboarding_topic	Fully replicated	4	1	0 B
adminPortal_sms_notification_topic	Fully replicated	4	1	795 KiB
dev-connect-cluster-configs	Fully replicated	1	0	19.9 KiB
dev-connect-cluster-offsets	Fully replicated	25	0	118 KiB
dev-connect-cluster-status	Fully replicated	5	0	537 KiB
eventtopics	Fully replicated	1	0	0 B
gst_invoice_download_topic	Fully replicated	4	0	0 B
gst_report_status_topic	Fully replicated	4	0	0 B
inb_offline_gateway_pooling_local	Fully replicated	4	1	0 B
local-process-bookingdetails-topic	Fully replicated	4	0	0 B
local-process-bulk-refund-topic	Fully replicated	4	0	0 B
local_adminPortal_email_notification_topic	Fully replicated	4	1	0 B
local_adminPortal_merchantOnboarding_topic	Fully replicated	4	0	0 B
local_adminPortal_merchantonboarding_topic	Fully replicated	4	1	0 B
local_adminPortal_sms_notification_topic	Fully replicated	4	1	10.9 KiB
local_admin_portal_email_notification_topic	Fully replicated	4	0	0 B
local_admin_sms_notification_topic	Fully replicated	4	0	0 B

Details of the topic is visible in the console

- Consumer group details
- Partitions
- Offset value
- Messages
- Configurations

Also you can see the message details

Message

Partition

3

Offset

2,002

Size ?

269 B

Timestamp (local)

May 2, 2026 at 2:59:35 PM GMT+5:30

Timestamp (UTC)

May 2, 2026 at 9:29:35 AM UTC

Unix timestamp

1777714175

Key format

Plain

Value format

Plain

Value

Key

Headers

{ "entityId": "f271f8b2-01b2-44f8-905c-61a4873dd..."

▼ {

entityId: "f271f8b2-01b2-44f8-905c-61a4873ddaf2",

entityType: "VIEW_ACCESS_OTP",

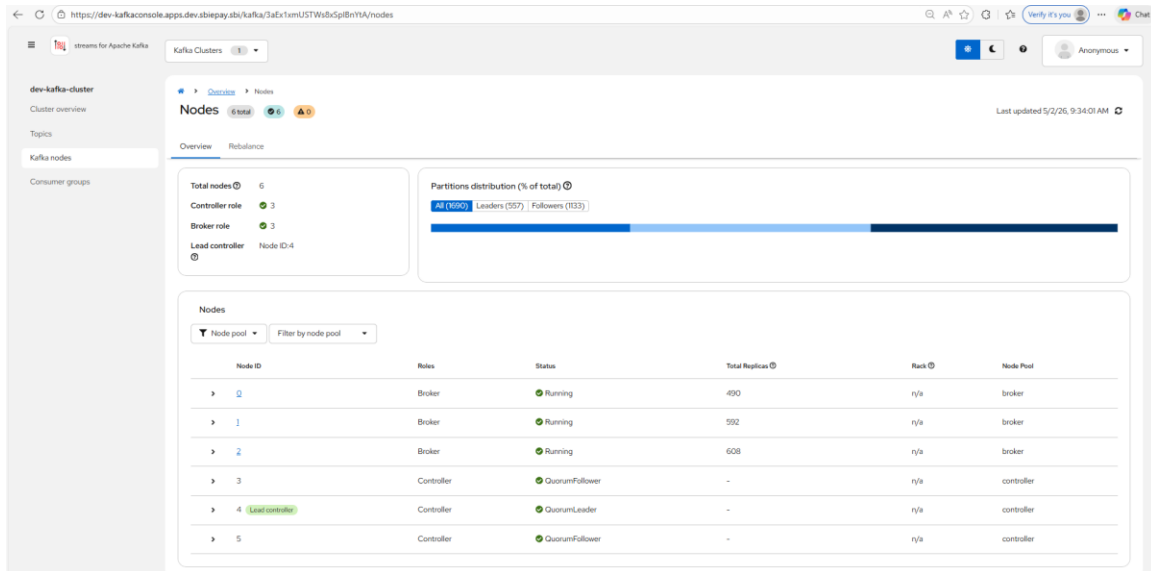
requestType: "SIGNUP",

mobileNumber: "9999988877",

message: "035948 is OTP for SBlePay login. OTP is valid for next 15 minutes. Do not share this OTP with anyone. - SBlePay"

}

Also you can see the Nodes details



You can see the Consumer group details as well

The screenshot shows the 'Consumer Groups' tab for the 'dev-kafka-cluster'. The left sidebar contains navigation links for 'Cluster overview', 'Topics', 'Kafka nodes', and 'Consumer groups'. The main content area displays a table of consumer groups with columns for Name, State, Overall lag, Members, and Topics.

Consumer group name	State	Overall lag	Members	Topics
admin-consumers	STABLE	0	6	local-adminPortal-email-notification-topic, adminPortal-args-notification-topic, local-adminPortal-args-notification-topic, 1 more
connect-netifi-aws-sink	STABLE	0	1	netifi-connect-kafka-sink
connect-netifi-aws-sink-dead	STABLE	0	1	netifi-connect-kafka-sink
gatewayOfflinePolicy-consumers	STABLE	1	4	id.offline-network-weather-local, gatewayOfflinePolicy-consumers-dev-args, gatewayOfflinePolicy-consumers-dev-args-2, 2 more
integration-consumers	STABLE	0	2	connect-weather-notification-topic, local-connect-weather-notification-topic
merchant-consumers	STABLE	19	25	aws-market-notification-topic, local-merchant-report-topic, local-adminPortal-merchantnotification-topic, 24 more

SOP Submitted by
Saadi M Muzzammil
Kafka Admin